Per-Site Retrieval of Deep Regolith Thermal Conductivity from the Restored Apollo 15 and 17 Heat-Flow Experiment: A Validation of the Global Model and Its Site-Level Error Budget

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May 22, 2026

Abstract

The deep thermal conductivity of lunar regolith, K_d, governs the subsurface equilibrium temperature, and through it the stability depth of polar volatiles and the heat budget of buried instruments. The conductivity model in routine use — Hayne et al. (2017) — was fitted to orbital brightness temperatures as a single Moon-wide average; how faithfully that global value represents any individual location has not been tested at depth, because the only in-situ deep-temperature records are the two Apollo Heat-Flow Experiment (HFE) boreholes. We use the restored 1971–1977 HFE record at Apollo 15 and 17 to carry out that test: a per-site retrieval of K_d obtained by integrating the 1-D heat equation under ephemeris-driven solar forcing, with shallow sensors excluded *a priori*. At the published basal heat flux, the deep-sensor misfit is minimised at $K^*_{d,\mathrm{A15}} = 4.86^{+1.12}_{-0.54}$ and $K^*_{d,\mathrm{A17}} = 11.23^{+8.12}_{-0.77}\,\mathrm{mW\,m^{-1}\,K^{-1}}$ (1σ non-parametric bootstrap) — values that bracket the global figure rather than reproduce it, and that we independently corroborate against a three-layer conductivity profile built from the Apollo core measurements. The two boreholes thus require distinctly different deep temperature gradients. Because that gradient depends only on the ratio Q_b/K_d, the difference can reside in the conductivity or in the basal heat flux Q_b; a pooled model comparison favours the conductivity interpretation ($\Delta\mathrm{AICc} \approx 5$ over a uniform-K_d alternative) but does not exclude a flux contrast. The practical conclusion is an *error budget*: applied at a specific site, a globally-averaged deep conductivity should be expected to depart from the local value by a factor of order two. We quantify that discrepancy here so that it can be carried, rather than ignored, in subsequent site-resolved thermal modelling.

Plain Language Summary. *How well the Moon's interior is insulated from the daily surface heating depends largely on the thermal conductivity of the compacted regolith — the broken-up rock — a metre or more down. No instrument in orbit can measure that deep property directly, so the conductivity model used across lunar science is a single value averaged over the whole Moon. An average is useful, but it raises a fair question: how far off is it at any one place? The only way to check is with temperatures measured in the ground, and those exist for just two locations — thermometers the Apollo 15 and 17 astronauts buried more than two metres deep in 1971–1972, whose full records were recovered from the original mission tapes only recently. We use those records to retrieve the deep conductivity at each of the two sites and find that the two values differ by roughly a factor of two, with the global average sitting between them rather than matching either. A complication is that the buried thermometers constrain a combination of conductivity and the heat escaping from the Moon's interior, not conductivity alone; weighing the possibilities, the difference is more likely a real conductivity contrast, though a difference in interior heat flow cannot be ruled out. The practical message is a number to carry forward: a globally-averaged conductivity, used at a specific site, can be wrong by about a factor of two — an error that polar-ice and instrument-design calculations should account for rather than overlook.*

Key Points

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- Per-site retrieval from the Apollo 15 and 17 Heat-Flow Experiment gives deep conductivities that bracket, rather than match, the global model value.
- The two retrieved values differ by about a factor of two and are corroborated by an Apollo-core-grounded three-layer model.
- A global deep conductivity applied at a single site should be expected to err by a comparable factor.

1 Introduction

Surface and shallow-subsurface temperatures of airless bodies are controlled by the thermal conductivity, density, and heat capacity of the upper few metres of regolith. For the Moon, the conductivity parameterisation in routine use is that of Hayne et al. (2017), who fitted a single $K(T, z)$ profile to the bolometric brightness temperatures of the Lunar Reconnaissance Orbiter Diviner radiometer (Paige et al., 2010; Vasavada et al., 2012). That profile is, by construction, a Moon-wide average: it is the deliberate and appropriate choice when a model must span the entire surface. It is now applied without modification to interpret Diviner-derived rock abundances, to calculate polar volatile stability, and to size buried-instrument heat budgets.

A global average invites a specific and answerable question: how closely does it match the conductivity at any one location? The answer matters because the quantities that depend on it — the depth to which polar ice is stable, the equilibrium temperature of a buried probe — respond to the *local* value, not the planetary mean. For the shallow regolith, orbital coverage allows the question to be addressed map-wide. For the *deep* regolith below about one metre, it cannot: orbital radiometry senses only the diurnal skin, and the sole temperature measurements made *within* the deep regolith are the two Apollo Heat-Flow Experiment (HFE) boreholes at Apollo 15 and 17 (Langseth et al., 1976; Nagihara et al., 2018). Those two records are therefore the only available test of how well a globally-averaged deep conductivity performs at a real site.

This paper is that test. We take the Hayne et al. (2017) functional form, hold it fixed, and retrieve the single deep-conductivity parameter K_d separately at each borehole from the restored 1971–1977 HFE record — a per-site validation rather than a global fit. The exercise has a deliberate downstream purpose: by measuring how far the per-site values depart from the global one, and how precisely each site can be pinned down, it yields a quantified *error budget* for the use of any global deep-conductivity model at a specific location.

Three features distinguish the analysis from earlier model–data comparisons.

(i) **A controlled, single-parameter per-site retrieval.** Earlier work evaluated published parameterisations as yes/no fits (Vasavada et al. 2012; Hayne et al. 2017; Nagihara et al. 2018) or jointly tuned several parameters of an alternative shape (Martínez and Siegler 2021). Here the entire Hayne et al. (2017) form is held fixed and only K_d is varied, so the two sites are compared on a single, interpretable axis.

(ii) **An independent, Apollo-grounded cross-check.** The retrieved K_d is corroborated against a discrete three-layer conductivity profile constructed from the Apollo core measurements themselves (§2), so the result does not rest on the Hayne functional form alone.

(iii) **Honest uncertainty propagation.** A non-parametric bootstrap ($N_{\mathrm{boot}} = 1500$) and an *a priori*, physics-based exclusion of the borestem-contaminated shallow sensors together ensure the reported error budget is neither understated nor circularly tuned.

An ephemeris-accurate solar-forcing chain (JPL DE440 lunar and solar positions) underlies the analysis; it removes the $\sim$1 lunar-hour drift of sinusoidal local-solar-time approximations and is required for the phase-folded diurnal-amplitude profile that defines the borestem cut (§3.2). The remaining sections describe the data and methods (§2), the per-site retrieval and its robustness (§3), the interpretation and its limits (§4), and the conclusions.

2 Data and Methods

2.1 Apollo HFE deep-sensor temperatures

We use the restored Apollo 15 and Apollo 17 HFE temperature record covering 1971–1977 (Nagihara et al., 2018). For every sensor we extract a multi-lunation *stability window* where the depth-mean temperature drift is $|d\langle T\rangle/dt| < 0.08\,\mathrm{K/yr}$ ($\approx 2.2 \times 10^{-4}\,\mathrm{K/d}$), and report the window-mean equilibrium temperature $T_{\mathrm{eq}} = \langle T(z)\rangle_{\Delta t}$ together with its within-window standard deviation. Sensors with central depth $z < 80\,\mathrm{cm}$ (the borestem-contaminated zone, see §3.3) are plotted but excluded from RMSE statistics. Figure 1 shows the full per-probe temperature record at each site, with the documented disturbance intervals shaded (Nagihara et al., 2018), the selected stability windows marked, and the fitted long-term drift $d\langle T\rangle/dt$ reported per sensor.

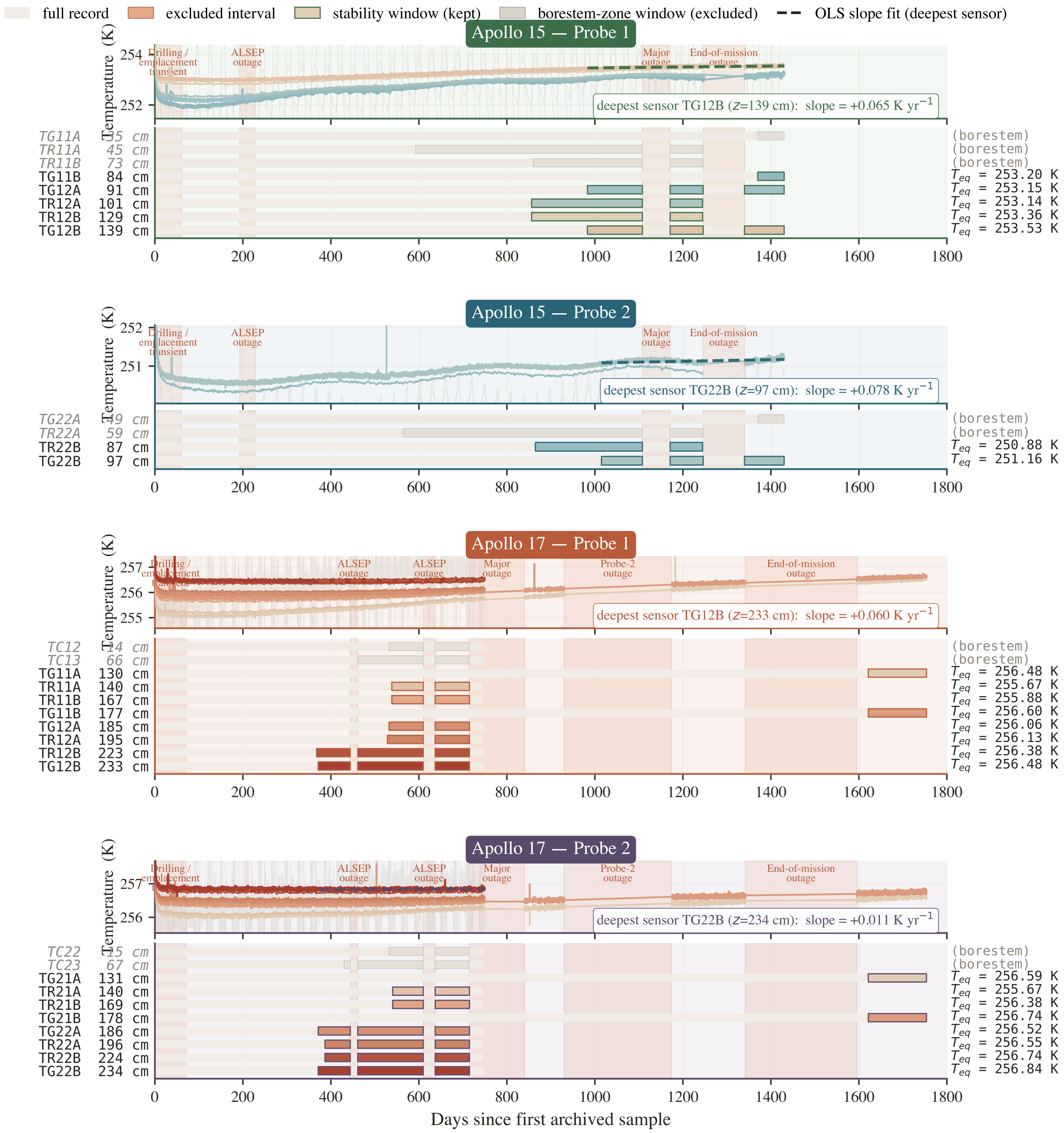

Figure 1. Per-probe stability-window timeline at Apollo 15 and 17 (4 probes). Each probe panel shows, above, the raw HFE traces with coral bands marking documented data-quality events and the selected stability windows outlined; and, below, a depth-coded per-sensor Gantt strip broken at outages. Selection criteria and the role of the window-mean T_{eq} are described in §2.

2.2 Surface solar geometry and incident insolation

We propagate every Apollo Unix timestamp through SPICE (Acton, 1996; Acton et al., 2018) using the JPL DE440 planetary ephemeris and the IAU-defined mean-Earth/polar-axis lunar body-fixed reference frame, with light-time and stellar-aberration corrections applied, and obtain the solar zenith angle $\theta_\odot(t)$ at each site. Because the Moon has no atmosphere, the solar irradiance incident on the local horizontal at the surface is simply

$$F_\odot(t) = \max\big(0,\ S_0 \cos\theta_\odot(t)\big), \tag{1}$$

with $S_0 = 1361\,\mathrm{W/m^2}$ (Kopp and Lean, 2011).

2.3 1-D solver and Hayne et al. (2017) thermophysics

Define the depth coordinate z as positive downward, with the regolith surface at $z = 0$ and the bottom of the modelled column at $z = z_{\max} = 5\,\mathrm{m}$. Conservation of energy in 1-D gives the heat equation

$$\rho(z)\, c_p(T)\, \frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\Big[\, K(T,z)\, \frac{\partial T}{\partial z}\Big]. \tag{2}$$

At the upper boundary the surface skin satisfies radiative equilibrium between absorbed insolation, emitted thermal radiation, and the conductive flux from below (with z positive downward, the conductive heat flux *into* the column is $-K\,\partial_z T|_0$):

$$(1 - A)\, F_\odot(t)\ -\ \varepsilon\sigma T_s^4\ -\ \big(-K\,\partial_z T\big)_{z=0}\ =\ 0, \tag{3}$$

solved iteratively for T_s at every step. At the lower boundary we impose a constant geothermal flux,

$$-K\,\partial_z T\big|_{z_{\max}} = Q_b. \tag{4}$$

The PDE is discretised on a geometric grid ($\Delta z_0 = 2\,\mathrm{mm}$, growth ratio 1.08) and integrated with a Crank–Nicolson scheme at $\Delta t = 1\,\mathrm{h}$, spun up for 30 lunations to $\max|\Delta T| < 5 \times 10^{-2}\,\mathrm{K}$. The Hayne et al. (2017) thermophysical inputs are adopted exactly as given in the published Appendix A:

$$K(T,z) = \Big\{K_s + (K_d - K_s)\big[1 - e^{-z/H}\big]\Big\} \cdot \Big[1 + \chi (T/T_{\mathrm{ref}})^3\Big], \tag{5}$$

$$\rho(z) = \rho_d - (\rho_d - \rho_s)\, e^{-z/H}, \tag{6}$$

$$c_p(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4, \tag{7}$$

with $K_s = 7.4 \times 10^{-4}\,\mathrm{W/(m\,K)}$, $K_d = 3.4 \times 10^{-3}\,\mathrm{W/(m\,K)}$, $H = 0.06\,\mathrm{m}$, $\chi = 2.7$, $T_{\mathrm{ref}} = 350\,\mathrm{K}$ (the Hayne et al. (2017)-Appendix-A normalisation; Vasavada *et al.* (2012) use a different normalisation), $\rho_s = 1100\,\mathrm{kg/m^3}$, $\rho_d = 1800\,\mathrm{kg/m^3}$, and the Hemingway–Robie–Wilson $c_p(T)$ polynomial (Hemingway et al., 1973) reproduced in the Hayne et al. (2017) Appendix. Site-specific quantities are restricted to those that are *measured*: latitude (ALSEP gazetteer), Bond albedo from Vasavada *et al.* (2012)'s lat-band averages ($A_{\mathrm{A15}} = 0.131$, $A_{\mathrm{A17}} = 0.137$), broadband emissivity $\varepsilon = 0.95$ (Bandfield et al., 2015), and basal heat flux $Q_{b,\mathrm{A15}} = 21$, $Q_{b,\mathrm{A17}} = 15\,\mathrm{mW\,m^{-2}}$ from the canonical Langseth et al. (1976); Nagihara et al. (2018) reductions — acknowledging the Saito et al. (2007); Nagihara et al. (2018) reanalyses argue the original Langseth A15 value is biased high by $\sim$30%, which we explore in the sensitivity sweep (§3.4). All fixed parameters are collected in Table 1.

We compare the Hayne et al. (2017) retrieval against two further regolith conductivity models. The first is the genuine Martínez and Siegler (2021) model, whose conductivity $K(T,\rho) = (A_1\rho + A_2)\, k_{\mathrm{am}}(T) + (B_1\rho + B_2)\, T^3$ is fully determined by temperature and density — it has no free K_d to retrieve, so it enters the comparison as a single fixed prediction. Here $k_{\mathrm{am}}(T)$ is the low-temperature amorphous-conduction polynomial of Martínez and Siegler (2021), and we use their published coefficients without modification.

The second is a deliberately simple discrete 3-layer model introduced in this work (*not* from Martínez and Siegler 2021). It approximates the documented Apollo regolith structure with a radiative-dominated surface layer (0–2 cm; Langseth et al. 1976), a rapid-compaction transition

Table 1. Thermophysical input parameters for the three model architectures. Only K_d is varied in the retrieval (§2.4) for the Hayne and 3-layer forms; the Martínez & Siegler (2021) model is parameter-free. Where two values appear they are A15 / A17. All three share the Hemingway–Robie–Wilson c_p polynomial $c_p = c_0 + c_1 T + \ldots + c_4 T^4$, $(c_0, \ldots, c_4) = (-3.61,\ 2.74,\ 2.36 \times 10^{-3},\ -1.23 \times 10^{-5},\ 8.91 \times 10^{-9})\,\mathrm{J\,kg^{-1}\,K^{-n}}$ (Hemingway et al., 1973).

Symbol / Parameter	Value	Unit	Source
Hayne et al. (2017) smooth-exponential (Hayne et al., 2017)			
K_s	7.4×10^{-4}	$\mathrm{W\,m^{-1}\,K^{-1}}$	
K_d^{ref}	3.4×10^{-3}	$\mathrm{W\,m^{-1}\,K^{-1}}$	
H	0.06	m	
χ	2.7	—	
T_{ref}	350	K	
ρ_s / ρ_d	1100 / 1800	$\mathrm{kg\,m^{-3}}$	
Martínez & Siegler (2021), T, ρ-dependent (Martínez and Siegler, 2021)			
Parameter-free: $K(T, \rho)$ is fully fixed by the Woods-Robinson amorphous-conduction polynomial and the density profile (no K_d to retrieve). Verified against the published code.			
This-work discrete 3-layer model			
K_s	7.4×10^{-4}	$\mathrm{W\,m^{-1}\,K^{-1}}$	Hayne et al. (2017)
z_1 / z_2	0.02 / 0.20	m	Langseth+ 1976
ρ_d (A15 / A17)	1825 / 1960	$\mathrm{kg\,m^{-3}}$	Langseth et al. (1976)
Site-specific measured quantities (all models)			
Latitude ϕ	26.1° / 20.2° N	—	ALSEP gazetteer
Bond albedo A	0.131 / 0.137	—	Vasavada et al. (2012)
Emissivity ε	0.95	—	Bandfield et al. (2015)
Basal heat flux Q_b	21 / 15	$\mathrm{mW\,m^{-2}}$	Langseth et al. (1976); Nagihara et al. (2018)
Solar constant S_0	1361	$\mathrm{W\,m^{-2}}$	Kopp and Lean (2011)

(2–20 cm), and a compacted deep layer. The 20 cm transition base is the depth at which the Hayne et al. (2017) exponential ($H = 6$ cm) has reached $\sim 97\%$ of K_d, so the boundary is consistent with the Hayne et al. (2017) comparison. Its only site-specific input is the deep bulk density — $\rho_d \approx 1825\,\mathrm{kg/m^3}$ at A15 and $1960\,\mathrm{kg/m^3}$ at A17 (Langseth et al., 1976) — which sets the transition-ramp curvature (a denser column compacts faster). As in the Hayne et al. (2017) form, K_d itself is the single swept free parameter, so the 3-layer model provides a direct test of how sensitive the retrieved K_d^* is to the assumed transition shape.

2.4 Per-site K_d retrieval (Hayne shape held fixed)

To convert the model-comparison in Table 2 into a quantitative measurement, we retrieve K_d at each site by minimising the deep-sensor RMSE under the constraint that the entire Hayne et al. (2017) functional form (eqs. 5–7) is otherwise unchanged. Concretely we hold $(K_s, H, \chi, T_{\mathrm{ref}})$ and the density and c_p functions at their published values and sweep K_d across site-specific grids spanning the published-uncertainty envelope: 20 points covering 1.5–$9.0 \times 10^{-3}\,\mathrm{W/(m\,K)}$ at A15 and 24 points covering 3.0–$18.0 \times 10^{-3}\,\mathrm{W/(m\,K)}$ at A17. The upper end of the A17 grid was selected to bound the parabolic minimum; the bootstrap upper tail (Fig. 5) extends slightly beyond the swept range by parabolic extrapolation. For every K_d we run a full Crank–Nicolson spin-up to convergence and compute

$$\mathrm{RMSE}(K_d) = \left\langle \left[T_{\mathrm{model}}(z_i; K_d) - T_{\mathrm{eq}}^{\mathrm{HFE}}(z_i)\right]^2 \right\rangle_{z_i \geq 80\,\mathrm{cm}}^{1/2}. \tag{8}$$

The point estimate K_d^* is the parabolic minimum through the three grid values bracketing arg min RMSE. Its uncertainty is quantified by a non-parametric bootstrap with $N_{\mathrm{boot}} = 1500$ resamples of the deep-sensor set drawn with replacement, with the ± 2.5 cm sensor-placement uncertainty (Nagihara et al., 2018) propagated as a per-resample Gaussian depth jitter, following standard procedure for small-sample regression (Press et al., 2007). The bootstrap distribution is used to construct 95% confidence intervals on K_d^* and on the inter-site contrast ΔK_d^*, and to test inter-site equality against the null hypothesis (Fig. 5). We report percentile-method intervals throughout; bias-corrected

Table 2. Deep-sensor ($z > 80$ cm) validation statistics for the three conductivity models. K_d in mW m^{-1} K^{-1} (95% bootstrap CI in brackets); N is the deep-sensor count; bias = mean(model−obs) in K; ΔAICc is relative to the best model at each site. The first row per site holds K_d at the published global value; the lower two retrieve it.

Site	Model	K_d	N	RMSE	Bias	ΔAICc
Apollo 15	Hayne (2017), global	3.4 (fixed)	7	**1.37**	+1.03	+1.7
Apollo 15	Hayne shape, K_d^* fit	4.86 [3.94, 14.36]	7	**0.90**	+0.00	**0.0**
Apollo 15	3-layer, K_d^* fit	5.70	7	**0.90**	−0.02	+0.0
Apollo 17	Hayne (2017), global	3.4 (fixed)	16	**2.96**	+2.88	+70.5
Apollo 17	Hayne shape, K_d^* fit	11.23 [10.18, 21.62]	16	**0.30**	−0.03	**0.0**
Apollo 17	3-layer, K_d^* fit	11.27	16	**0.39**	−0.04	+8.5

accelerated (BCa) intervals (Efron and Tibshirani, 1993) are within 1.3 mW/(m K) of the percentile endpoints at both sites and on the contrast, and do not change any qualitative conclusion. The retrieval has exactly one free parameter.

To compare models with different free-parameter counts on the same deep-sensor set we use the small-sample-corrected Akaike information criterion, AICc $= N \ln(\mathrm{RSS}/N) + 2k + 2k(k+1)/(N-k-1)$, where RSS is the deep-sensor residual sum of squares and k is the number of fitted parameters ($k = 0$ for the fixed-K_d rows, $k = 1$ for each retrieval). We report differences ΔAICc relative to the best model at each site; by convention ΔAICc $\lesssim 2$ indicates statistically indistinguishable models and $\gtrsim 10$ a decisive preference. We do not report a reduced χ^2: the within-stability-window scatter ($\sigma \sim 0.03$–0.05 K) measures short-term thermometer precision rather than the model-adequacy error budget, so it yields $\chi^2_\nu \gg 1$ for every model and is uninformative for model selection here.

3 Results

3.1 Annual-mean subsurface profile

Fig. 2 compares three modelled time-averaged $\langle T(z) \rangle$ profiles — the Hayne et al. (2017) smooth-exponential form at its published global K_d, the Martínez and Siegler (2021) T, ρ-dependent model, and the this-work discrete 3-layer model — against the Apollo HFE stability-window means at both sites. With *no per-site tuning*, the published global Hayne et al. (2017) value ($K_d = 3.4$ mW m^{-1} K^{-1}) under-predicts the deep gradient at both sites (Table 2), and neither published model reproduces both records simultaneously. The shaded band ($z < 80$ cm, the borestem-contaminated zone) marks sensors that are excluded *a priori* from the quantitative comparison; see §3.3.

3.2 Diurnal-amplitude depth profile

The mean profile in Fig. 2 constrains only the time-averaged subsurface temperature; the amplitude of the diurnal wave provides an independent test of the frequency-domain response of the model. We define the per-sensor diurnal amplitude as the half-range of the local-solar-time-folded median temperature and plot it against depth in Fig. 3. Above $z = 80$ cm the observed amplitudes lie an order of magnitude or more above the modelled regolith attenuation curve, consistent with the borestem heat-short identified by Nagihara et al. (2018) (§3.3). Below 80 cm both models predict an amplitude $\lesssim 10^{-7}$ K (corresponding to $\sim$18 e-foldings of the diurnal skin depth δ for $\delta \sim 3$–5 cm), two to three orders of magnitude below the digitisation noise floor of the restored 1971–1977 record ($\sim$0.05–0.2 K). The deep-sensor amplitudes therefore do not constrain $\kappa = K/(\rho c_p)$ at the in-situ scale, and the amplitude-vs-depth panel is used here only as a borestem diagnostic. Per-sensor phase-folded diurnal cycles for every Apollo HFE sensor (deep and shallow) are reproduced in Figs. S1–S2 of the SI.

3.3 Borestem heat-short artefact

The shallow-sensor amplitude excess in Fig. 3 is the same anomaly reported by Nagihara et al. (2018) and attributed to heat conducted along the fibreglass borestem from the lunar surface, which thermally short-circuits the upper $\sim$80 cm of regolith. We use this independent diagnostic — the

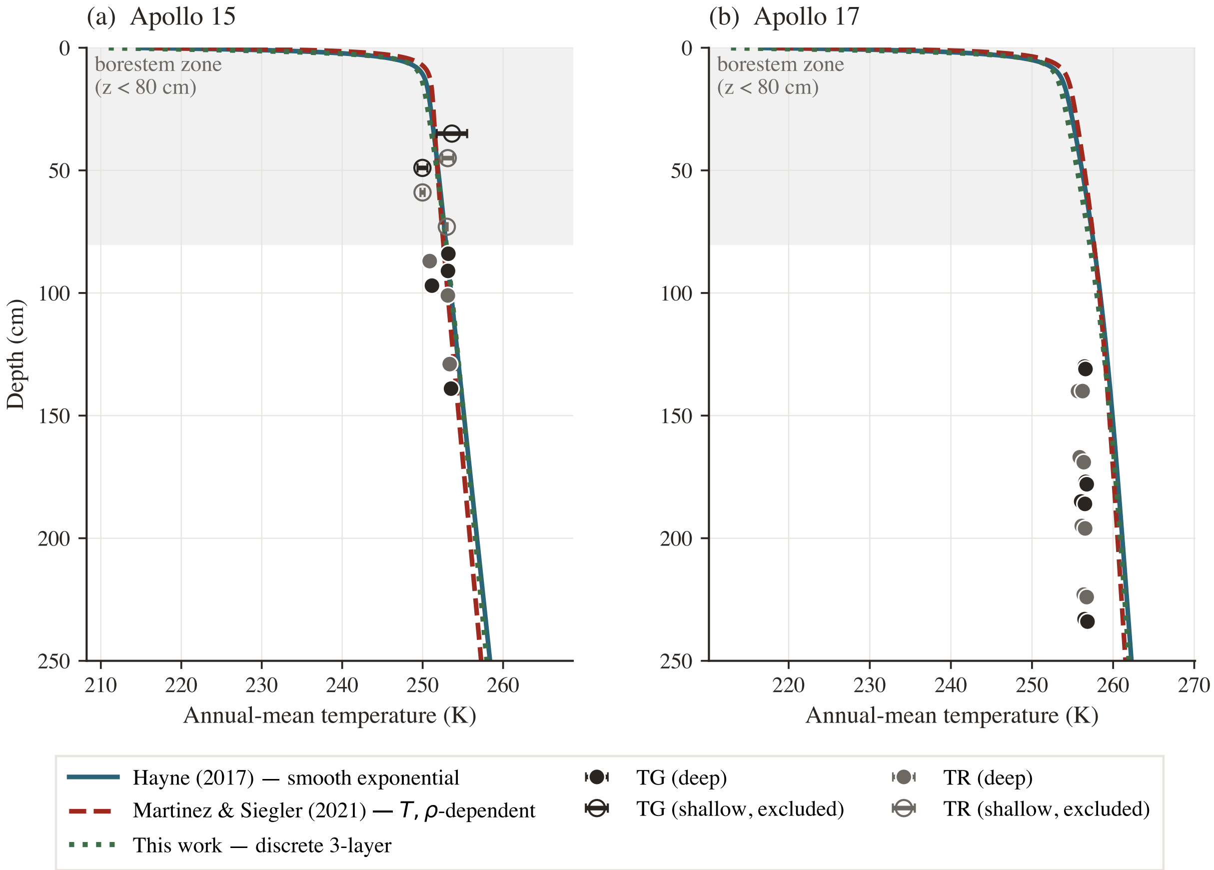

Figure 2. Annual-mean subsurface temperature $\langle T(z)\rangle$ for three conductivity models at their nominal (un-retrieved) parameters — Hayne et al. (2017) smooth-exponential (solid teal, global $K_d = 3.4$), Martínez and Siegler (2021) (dashed red), this-work 3-layer (dotted green) — against the Apollo HFE stability-window means. Markers are coded by sensor type (charcoal = TG, grey = TR); horizontal bars are the within-window standard deviation. The shaded band is the borestem-contaminated zone ($z < 80$ cm), excluded from the RMSE.

amplitude-vs-depth profile, evaluated against an amplitude-attenuation prediction derived only from $K(T, z)$, $\rho(z)$, $c_p(T)$ — to define the deep ($z > 80$ cm) validation set used in Table 2. The exclusion is therefore not based on the residuals it produces.

3.4 Robustness to auxiliary parameter uncertainties

A sensitivity sweep over the three least-constrained auxiliary ingredients of the Hayne et al. (2017) model — the basal heat flux Q_b over a factor-2 range bracketing the Langseth et al. 1976/Saito et al. 2007; Nagihara et al. 2018 debate, the radiative coefficient χ over the full Hayne et al. (2017)/ Vasavada *et al.* (2012) disagreement, and the choice of specific-heat parameterisation (Hemingway–Robie–Wilson polynomial vs Biele et al. 2022 rational fit) — shifts the deep-sensor RMSE by < 0.5 K at both sites; the component-by-component K_d^* error budget is tabulated in Table 3. K_d is treated separately as the controlled retrieval parameter in §3.5.

3.5 Per-site K_d retrieval

Fig. 4 shows the deep-sensor RMSE (eq. 8) as a function of K_d at each landing site, with the Hayne et al. (2017) functional form (eqs. 5–7) otherwise unchanged. Both curves have a clean, well-resolved minimum interior to the swept range. Extending the sweep well beyond the fitting grid confirms that each minimum is a genuine bracketed bowl rather than a descending shoulder: the deep-sensor RMSE rises monotonically on the high-K_d side at both sites (Fig. 4). A longer sweep would therefore only *raise* the RMSE and cannot relocate either minimum. The parabolic point estimates with

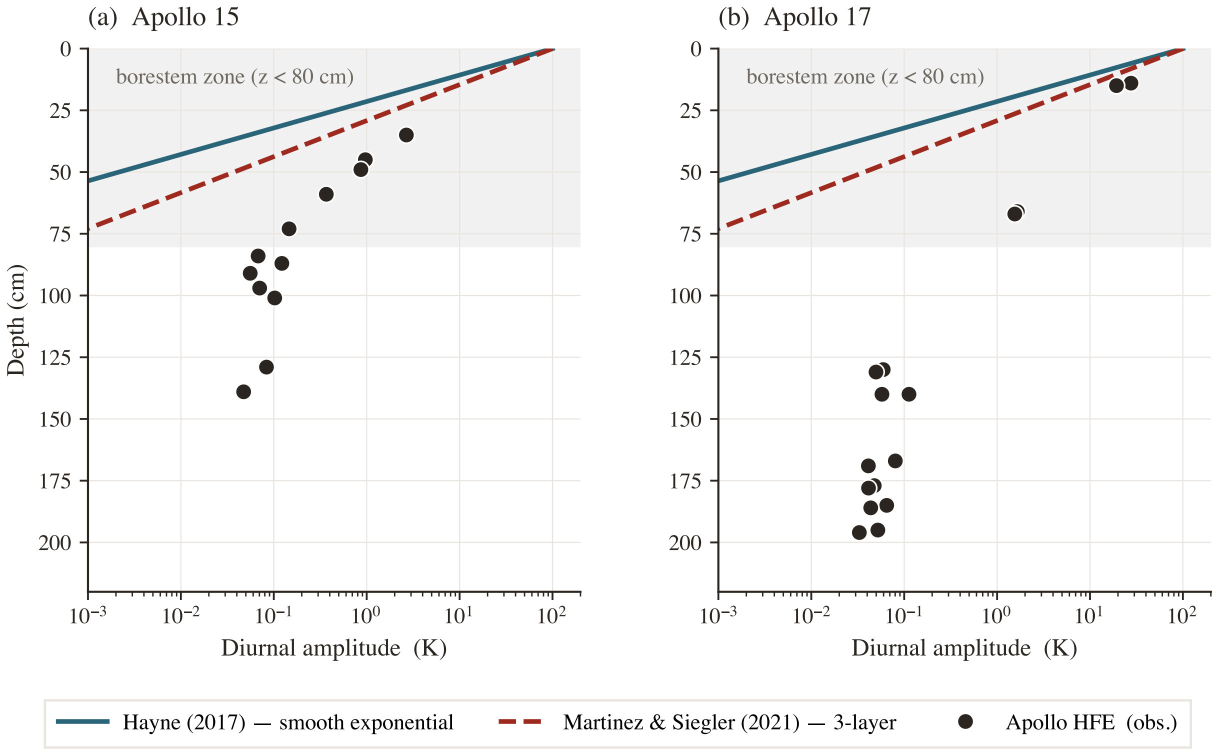

Figure 3. Diurnal amplitude (half-range of the local-solar-time-folded median temperature) versus sensor depth at both Apollo sites. Black markers: observed; navy curve: Hayne et al. (2017) reference; grey curve: 3-layer alternative. The shaded band is the borestem zone ($z < 80$ cm). The interpretation of the amplitude excess as the borestem signature is given in §3.3.

asymmetric 1σ bootstrap uncertainties (and 95% intervals in square brackets) are

$$K^*_{d,\mathrm{A15}} = 4.86^{+1.12}_{-0.54}\,[3.94,\ 14.36]\,\mathrm{mW/(m\,K)}, \qquad K^*_{d,\mathrm{A17}} = 11.23^{+8.12}_{-0.77}\,[10.18,\ 21.62]\,\mathrm{mW/(m\,K)},$$

with corresponding deep-sensor RMSE $= 0.90$ K ($N = 7$) and 0.30 K ($N = 16$) respectively (Fig. 5). The retrieval improves on the published global-K_d Hayne et al. (2017) baseline at both sites (Table 2: $1.37 \to 0.90$ at A15, $2.96 \to 0.30$ at A17), and the bias collapses to ≤ 0.03 K, indicating that K_d is the principal identified lever between the Hayne et al. (2017) form and the in-situ HFE record once the borestem zone is excluded. The small-sample-corrected Akaike statistic confirms this: the per-site retrieval is decisively preferred over the fixed global value at A17 (ΔAICc $= 70.5$) and weakly preferred at A15 (ΔAICc $= 1.7$; Table 2). Repeating the retrieval under the discrete 3-layer model returns $K^*_d = 5.70$ (A15) and 11.27 (A17) $\mathrm{mW\,m^{-1}\,K^{-1}}$. The two $K(z)$ shapes are statistically indistinguishable at A15 (ΔAICc $= 0.04$); at A17 the smooth-exponential form fits better (ΔAICc $= 8.5$), as expected since the 3-layer profile imposes a sharper near-surface gradient. Crucially, the *retrieved* K^*_d agrees between the two shapes to within 1% at A17 and 17% at A15 — both well inside the bootstrap intervals — so the inter-site contrast itself is a property of the data and not an artefact of the assumed transition shape, even though the two shapes are not equally good fits.

Sensitivity to the borestem exclusion depth. Because the retrieval uses only sensors below the 80 cm borestem cut, we re-ran it with the exclusion depth swept over 60–90 cm. The retrieved values are stable for every cut at or below the borestem zone: $K^*_{d,\mathrm{A15}} = 4.7$–$4.9$ and $K^*_{d,\mathrm{A17}} = 11.2\,\mathrm{mW/(m\,K)}$ for cuts of 70, 80 and 90 cm, giving a contrast of 6.3–6.5 mW/(m K) — unchanged within rounding from the nominal value. Only the shallowest, 60 cm cut deviates ($K^*_{d,\mathrm{A17}} = 18.8$, contrast 14.2): at that depth a sensor inside the borestem-affected zone re-enters the A17 set and inflates the retrieved gradient, which is precisely the contamination the amplitude-based cut (§3.3) is designed to exclude. The result is thus insensitive to the exclusion depth across its physically defensible range, and the 60 cm outlier confirms rather than undermines the borestem diagnostic.

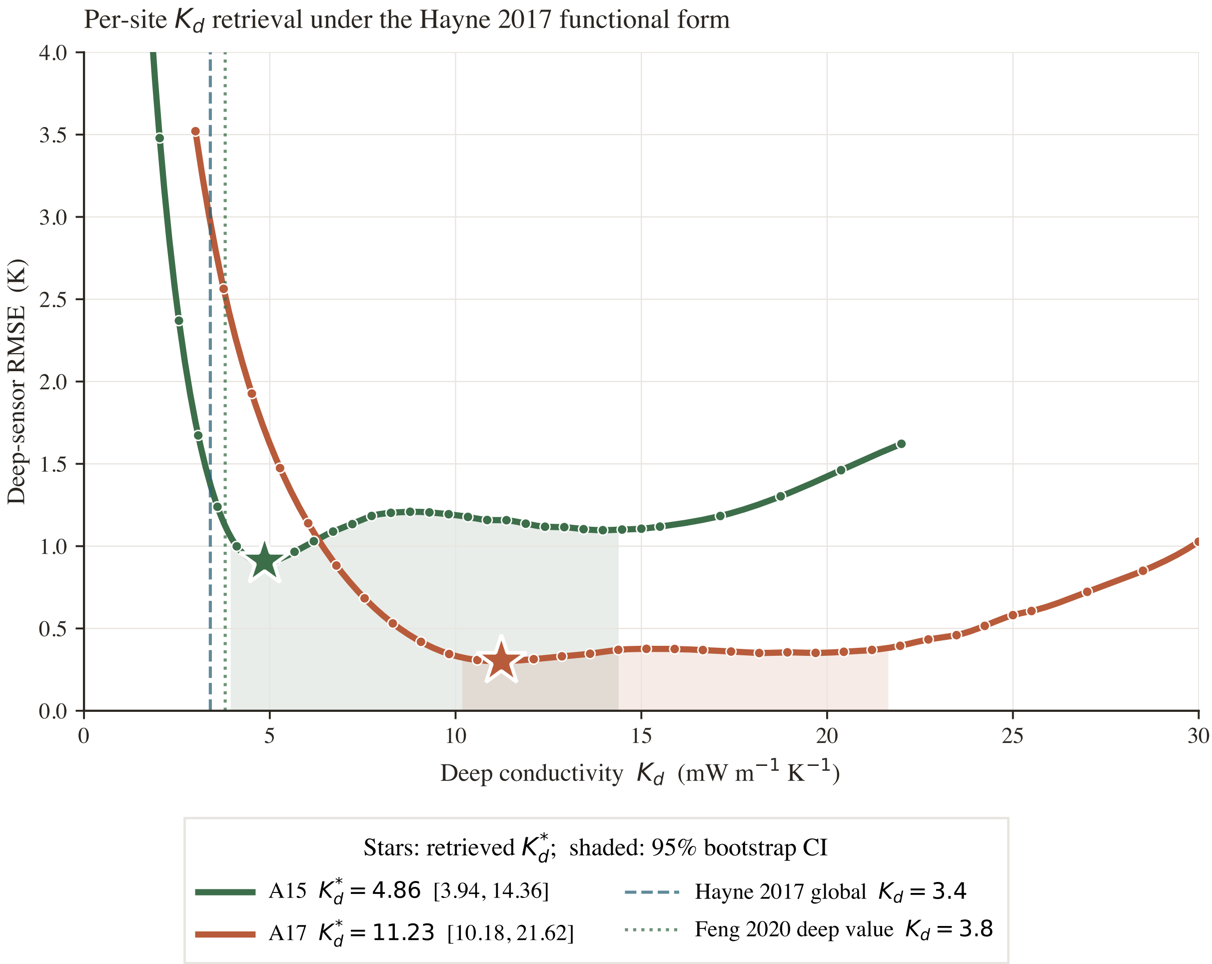

Figure 4. Per-site K_d retrieval under the Hayne et al. (2017) functional form, with all other Hayne et al. (2017) parameters held at their published values. Deep-sensor RMSE(K_d) at each site, extended past the fitting grid so each curve is shown rising again on the high-K_d side: the minima are genuine bracketed bowls, not descending shoulders. Markers: forward-model evaluations; stars: retrieved K_d^*; shaded band: 95% bootstrap CI. Dashed vertical: Hayne et al. (2017) global $K_d = 3.4$; dotted: the Feng et al. (2020) deep value $K_d = 3.8$.

We report two distinct summary statistics, and keep them clearly separated. The per-site K_d^* values quoted above (4.86, 11.23 mW/(m K)) are the *parabolic point estimates* of each RMSE minimum; the bootstrap *medians* of the same quantities are 4.88 and 11.51 (Table 3), marginally higher because each bootstrap distribution is right-skewed. The inter-site contrast is computed as the bootstrap median of the paired difference $K^*_{d,\mathrm{A17}} - K^*_{d,\mathrm{A15}}$ *within each resample* — not as the difference of the two separate point estimates — and is $\Delta K_d^* = 6.63^{+7.43}_{-1.25}$ mW/(m K) (1σ, asymmetric) with a 95% bootstrap confidence interval $[-3.2, 16.6]$ mW/(m K) that just touches zero at its lower bound (Fig. 5; $p \approx 0.05$ for the null of inter-site equality with the extended A15 sweep grid). The paired construction is the statistically correct one — it preserves the resample-level correlation between the sites — and it is why ΔK_d^* (6.63) is the difference of the bootstrap medians (11.51 – 4.88) rather than of the point estimates (11.23 – 4.86 = 6.37). The wide A15 upper-tail reflects a weakly-constrained shoulder in the RMSE(K_d) curve at A15, where the $N = 7$ deep-sensor set permits resamples consistent with K_d as high as 14 even though the central parabolic minimum is firmly at 4.86. The reported 1σ intervals propagate sensor-set composition, sensor-placement depth uncertainty (± 2.5 cm, Nagihara et al. 2018) and the parabolic curve-fit error at the bootstrap minimum; the other Hayne et al. (2017) parameters (K_s, H, χ, ρ, c_p) are held at their published values, and the effect of varying them within their published ranges is tabulated as a per-component error budget in Table 3. Both retrievals reject the global value $K_d = 3.4$ mW/(m K) adopted by

Hayne et al. (2017) at the 95% level.

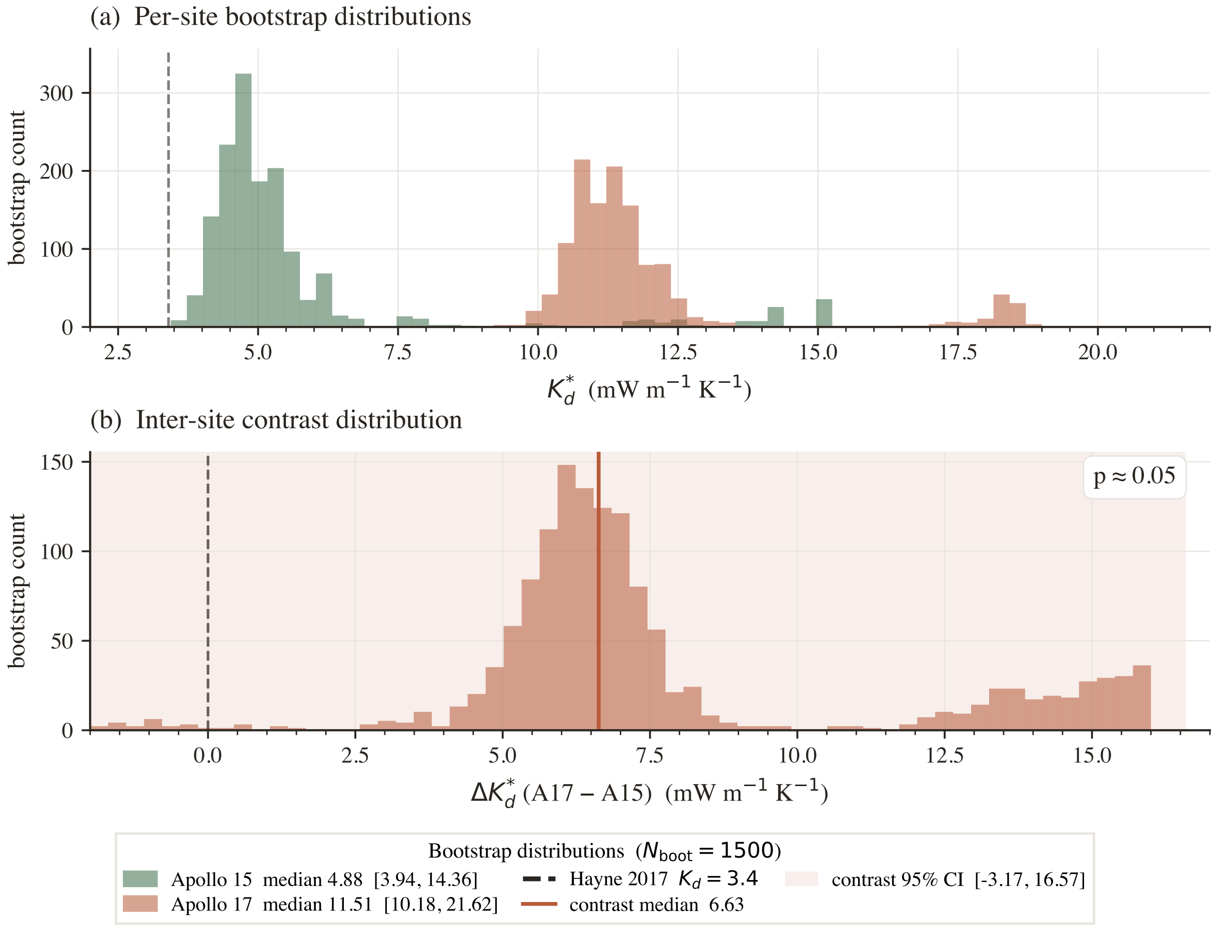

Figure 5. Bootstrap distributions ($N_{\mathrm{boot}} = 1500$, deep-sensor resampling with ± 2.5 cm depth jitter). (a) Per-site K_d^* histograms with 95% CIs in the legend; dashed vertical at the Hayne et al. (2017) global value. (b) Inter-site contrast $\Delta K_d^* = K_d^{\mathrm{A17}} - K_d^{\mathrm{A15}}$; 95% CI shaded, lower bound just touching zero ($p \approx 0.05$).

Table 3. Per-component error budget for the retrieved K_d^*. Each row is the propagated 1σ shift in K_d^* from varying one input within its published uncertainty, evaluated at the bootstrap median; the components and how each is derived are described in §3.5. All values in mW m^{-1} K^{-1}.

Component	σ (A15)	σ (A17)
σ_{stat} (sensor composition + 2.5 cm depth jitter)	0.83	4.44
σ_{Q_b} (Saito–Nagihara $\pm 30\%$)	1.47	3.45
σ_{χ} (radiative $\chi \pm 30\%$, $\Delta K_d/K_d \approx 15\%$)	0.73	1.73
σ_H (data-driven, joint K_d–H sweep)	0.31	1.44
σ_{K_s} ($K_s \pm 2\%$)	0.10	0.23
σ_ρ ($\rho \pm 0.5\%$)	0.02	0.06
Total (quadrature sum)	**1.87**	**6.06**
Bootstrap median K_d^*	4.88	11.51

Each component of Table 3 is derived as follows. σ_{stat} is the half-range of the bootstrap 16–84 percentile band (sensor-set composition plus ± 2.5 cm depth jitter). σ_{Q_b} is the $\pm 30\%$ Saito–Nagihara basal-flux envelope mapped through the steady-state K_d/Q_b degeneracy. σ_χ is the chain-rule shift ($\Delta K_d/K_d \approx 15\%$) from a $\pm 30\%$ perturbation of the radiative coefficient χ. σ_H is the half-range of the per-H K_d^* across the joint 11×8 K_d–H sweep ($H = 1$–11 cm); σ_{K_s} and σ_ρ follow from $\pm 2\%$ and $\pm 0.5\%$ perturbations of K_s and ρ. The decomposition makes clear which terms dominate.

At Apollo 15 the budget is led by the basal-flux uncertainty $\sigma_{Q_b} \approx 1.5\,\mathrm{mW/(m\,K)}$ (driven by the $\pm 30\%$ spread in the Langseth–Saito–Nagihara reanalyses) and the statistical sensor-composition term $\sigma_{\mathrm{stat}} \approx 0.83$ (the $N=7$ small-sample bootstrap width plus the $\pm 2.5\,\mathrm{cm}$ sensor-placement jitter); the remaining four terms together contribute under $0.8\,\mathrm{mW/(m\,K)}$ in quadrature. At Apollo 17 the statistical term is itself the dominant component ($\sigma_{\mathrm{stat}} \approx 4.4$), reflecting the long upper shoulder of the A17 bootstrap distribution (Fig. 5a); σ_{Q_b} is second (~ 3.5), and the structural Hayne et al. (2017)-parameter uncertainties (χ, H, K_s, ρ) all sit at or below $2\,\mathrm{mW/(m\,K)}$.

Two qualitative implications follow. First, the absolute K_d^* at each site is bounded by Q_b and by the small-sample bootstrap, both of which are properties of the data (not of the Hayne et al. (2017) form); no further refinement of the regolith parameterisation will tighten these. Second, the inter-site contrast is far less sensitive to Q_b than either absolute value, because the K_d/Q_b degeneracy is a *global* scaling: a uniform rescaling of Q_b at both sites cancels exactly in ΔK_d^*. The contrast is therefore exposed to Q_b uncertainty only through the *difference* between the two sites' rescaling factors.

We test this explicitly by rescaling Q_b *independently* at the two sites across the full published envelope ($Q_{b,\mathrm{A15}} \in [14, 25]$, $Q_{b,\mathrm{A17}} \in [10, 18]\,\mathrm{mW/m^2}$; Fig. 6a). Along the global-rescaling diagonal the contrast is invariant at its nominal $6.6\,\mathrm{mW/(m\,K)}$. Off the diagonal it varies: the most adverse combination — A15 pushed to the top of its Q_b envelope and A17 to the bottom — shrinks the contrast to $\Delta K_d^* \approx 1.9\,\mathrm{mW/(m\,K)}$ ($\sim 0.4\,\sigma$), while the opposite extreme widens it to ~ 10.6. Crucially, the contrast *never reverses sign* anywhere within the published Q_b envelope: A17 remains the more conductive site under every admissible per-site basal-flux assignment. We therefore do not claim that the contrast is significant under *all* independent Q_b revisions — a sufficiently large site-specific revision (such as a -33% reduction of $Q_{b,\mathrm{A17}}$ unaccompanied by any change at A15) can reduce it below 1σ — but its *sign* is robust, and that sign is the basis for the central claim that no single globally-uniform K_d fits both records.

Consistency checks. Three diagnostics test the per-site K_d^* values; we are explicit about which is genuinely independent and which are internal. *(i)* Diviner surface-temperature closure — the one fully independent test, against an external dataset not used in the retrieval. The modelled surface trace at each site, run with the per-site retrieved K_d^*, reproduces the Diviner Lunar Radiometer Global Cumulative Product (Williams et al., 2017) diurnal shape with terminator timing, plateau width, and night-time floor faithfully captured at both sites (Fig. 7; full-cycle RMSE 12.4 K and 17.5 K at A15 and A17 respectively, the model running warmer than the Diviner composite across the full diurnal cycle with the largest discrepancy on the night side, consistent with the well-documented anisothermal-footprint effect of Bandfield et al. (2015); Hayne et al. (2017)). This is a substantial surface-level systematic, so its propagation into the deep-K_d retrieval must be quantified rather than assumed. We do so by perturbing the surface energy balance through the Bond albedo — the dominant lever on absolute surface temperature — by ± 0.04, four times the A15/A17 albedo uncertainty, and re-retrieving K_d at each perturbed albedo. The induced surface-temperature swing of $\sim 4.5\,\mathrm{K}$ (comparable to the closure RMSE itself) shifts K_d^* by $1.1\,\mathrm{mW\,m^{-1}\,K^{-1}}$ at A15 (23%) and 1.9 at A17 (17%). The deep retrieval is therefore *not* fully decoupled from the surface state; but these shifts are well within the bootstrap confidence intervals (which span a factor of 3.6 at A15), and across the entire perturbation range A17 remains the more conductive site. The surface-closure mismatch thus broadens the per-site uncertainty modestly but does not generate the inter-site difference: the closure test validates the diurnal *shape* and the ephemeris forcing chain, and the albedo-perturbation test bounds the residual surface-state sensitivity of K_d^*.

The remaining two diagnostics are *internal consistency* checks — they re-use the same boreholes and the same deep-sensor set, so they probe the stability of the retrieval rather than validate it against independent data. *(ii)* TG vs. TR cross-prediction: fitting K_d using only the gradient-bridge sensors (TG) and predicting the ring-bridge (TR) sensors, and *vice versa*, yields held-out RMSEs within $\sim 0.25\,\mathrm{K}$ of the in-sample values at both sites. *(iii)* Leave-one-deepest-out: withholding the deepest sensor ($z = 139\,\mathrm{cm}$ at A15, $z = 234\,\mathrm{cm}$ at A17) and refitting K_d on the remainder predicts the held-out temperature to within $|\Delta| = 0.31$ and $0.16\,\mathrm{K}$, respectively. Both confirm the retrieval

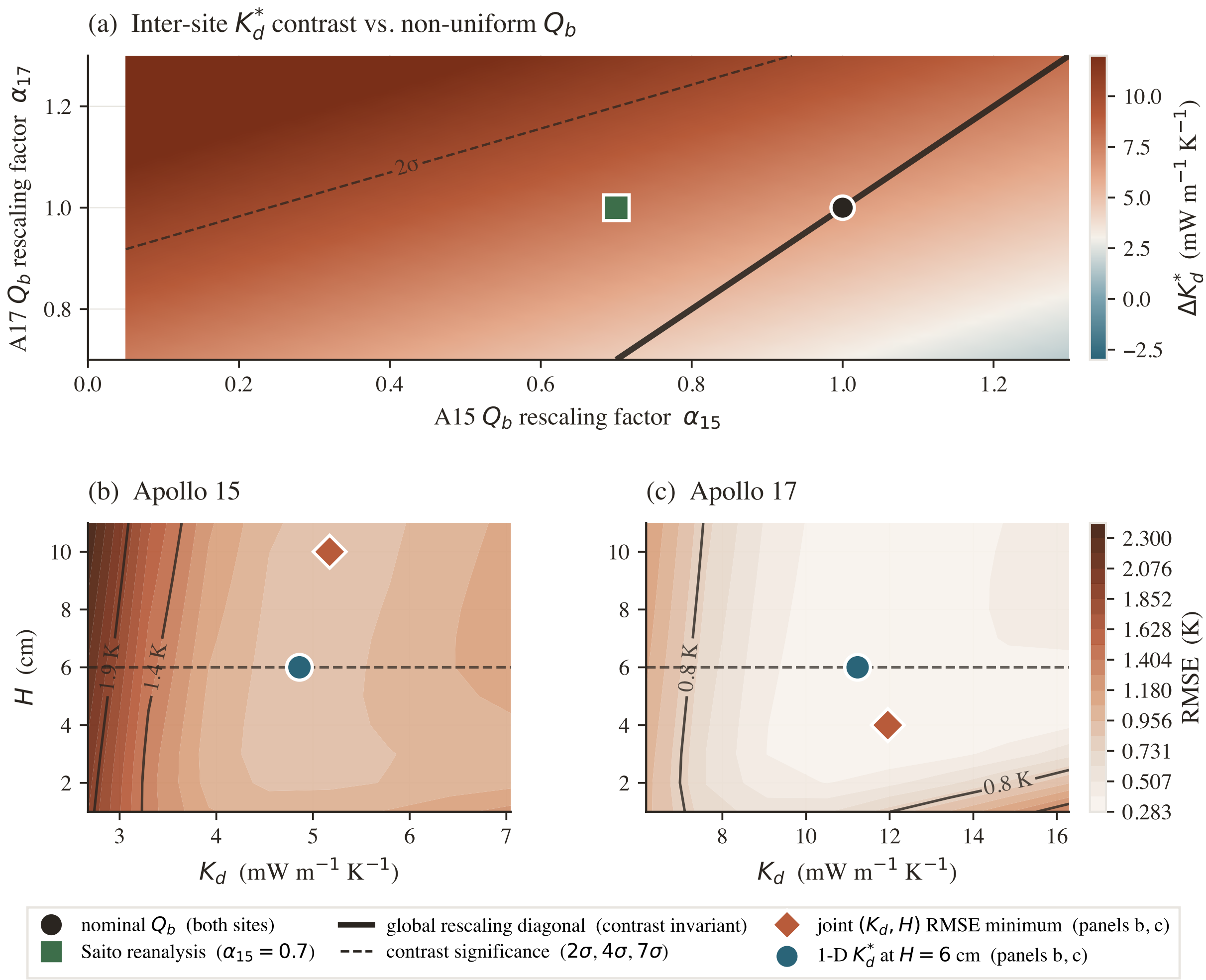

Figure 6. Robustness of the per-site K_d^* contrast. (a) Contrast ΔK_d^* as a function of per-site Q_b rescaling factors α_{15}, α_{17}. Solid diagonal: global rescaling (contrast invariant); dashed contours: bootstrap-σ significance; black circle: nominal Q_b; green square: Saito et al. (2007) A15 reanalysis. (b),(c) Joint (K_d, H) RMSE surface at each site over the swept $H = 1$–11 cm range; the coral diamond marks the joint RMSE minimum and the teal circle the 1-parameter retrieval at $H = 6$ cm. The RMSE surface is extremely flat in H — it varies by < 0.02 K across $H = 3$–11 cm — so H is effectively unconstrained, with shallow interior minima at $H \approx 10$ cm (A15) and 4 cm (A17). The retrieved K_d is nonetheless stable: the joint minimum and the 1-D retrieval agree in K_d to within the 0.5 K RMSE contour, confirming that K_d, not H, is the identified parameter.

is not driven by a single sensor or sensor type, but neither is an independent test of the inter-site result.

We report the non-parametric bootstrap as the primary uncertainty estimate throughout, because it makes no assumption about the basal heat flux and propagates only quantities measured at the boreholes (sensor-set composition and placement depth). As an independent cross-check on the *direction* of the contrast — not as a second estimate of its magnitude — we also ran a Bayesian inversion that propagates the Saito et al. (2007); Nagihara et al. (2018) prior on Q_b (Fig. 8). Its marginal MCMC medians are $K_{d,\mathrm{A15}} = 3.9$ and $K_{d,\mathrm{A17}} = 13.1$ mW/(m K) (contrast median +9.2). These differ from the bootstrap point estimates (4.86, 11.23; contrast +6.6) by design and not by disagreement: the Q_b prior slides each marginal along the K_d/Q_b degeneracy ridge (§4.1), so the two methods are *not* independent measurements of the absolute K_d and their central values should not be expected to coincide. What the two methods do agree on, and all that we draw from the MCMC, is the ordering: $P(K_d^{\mathrm{A17}} > K_d^{\mathrm{A15}}) \approx 96\%$, because the degeneracy ridge shifts both sites together and therefore preserves their difference. The bootstrap interval on the contrast magnitude ($\Delta K_d^* = 6.6$, 95% CI $[-3.2, 16.6]$) is the figure we carry forward.

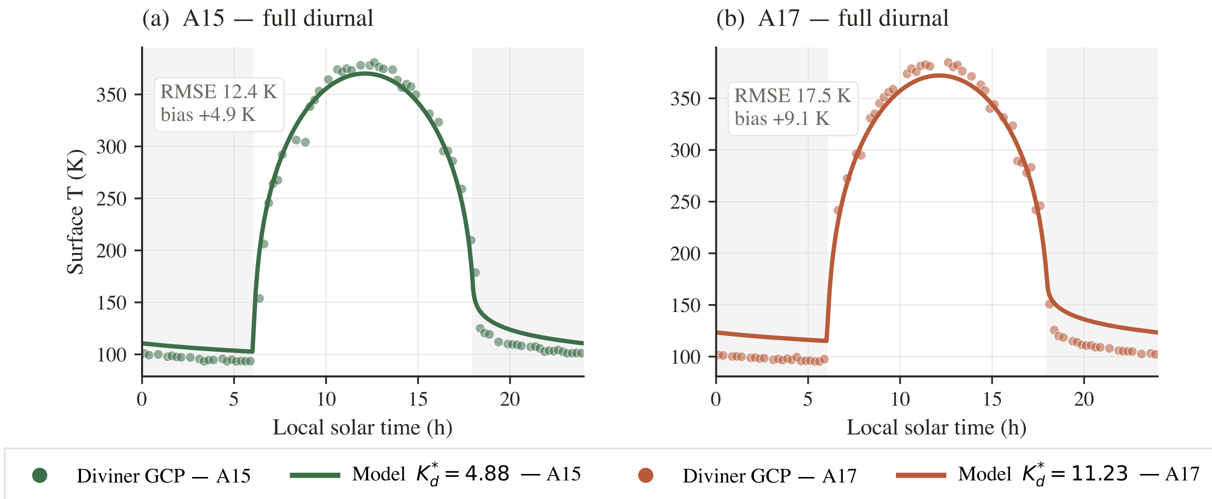

Figure 7. Surface-T closure against the Diviner Lunar Radiometer GCP. Modelled surface T (solid) at A15 (teal) and A17 (coral) with per-site K_d^*, vs. Diviner (Williams et al., 2017) diurnal composite (markers); shaded bands = night (LST < 6 or ≥ 18 h). Full-cycle RMSE 12.4 and 17.5 K. The model is slightly warmer than Diviner across the cycle, with the largest discrepancy on the night side, consistent with anisothermal-footprint and sub-pixel rock-population effects (Bandfield et al., 2015; Hayne et al., 2017).

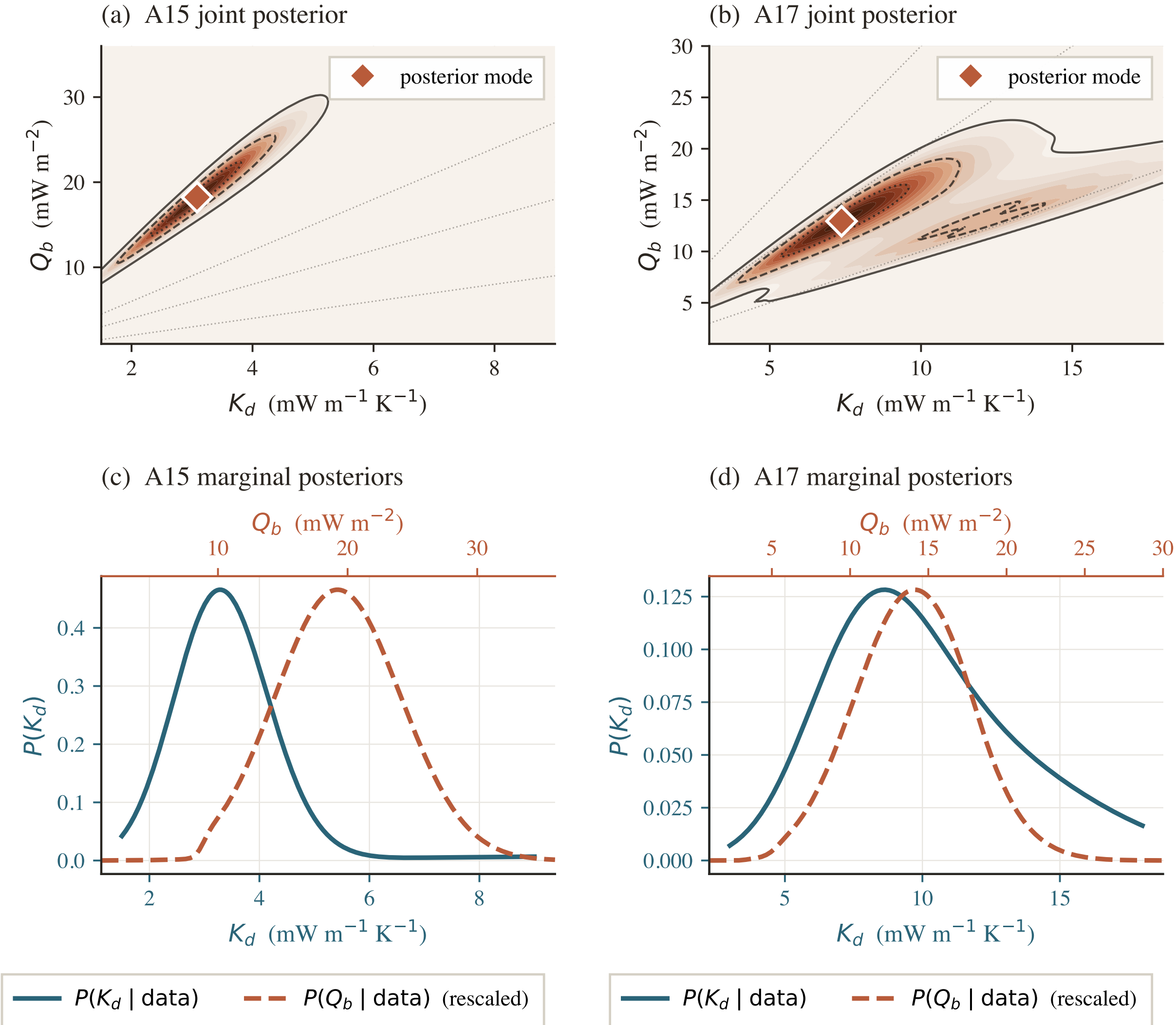

Figure 8. Joint (K_d, Q_b) posteriors and marginals with the Saito–Nagihara Q_b prior. (a),(b) Joint posterior on the (K_d, Q_b) plane; charcoal contours = 5/32/68% mass levels, coral diamond = mode, dotted rays = iso-ratio degeneracy $Q_b/K_d = \text{const}$. (c),(d) Marginals $P(K_d)$ (teal, bottom axis) and $P(Q_b)$ (coral dashed, top axis, rescaled). Medians: $K_{d,\text{A15}} = 3.9$, $K_{d,\text{A17}} = 13.1\,\text{mW}/(\text{m K})$; $P(K_d^{\text{A17}} > K_d^{\text{A15}}) \approx 96\%$.

4 Discussion

4.1 The two boreholes require different deep gradients

Under the Hayne et al. (2017) functional form at the published basal heat fluxes, the deep conductivity that reproduces the HFE record differs between the sites: $K_d^{\mathrm{A15}} = 4.86$ versus $K_d^{\mathrm{A17}} = 11.23\,\mathrm{mW/(m\,K)}$ (95% bootstrap CIs [3.94, 14.36] and [10.18, 21.62]). The contrast distribution (Fig. 5b) has its lower 95% bound just touching zero ($p \approx 0.05$): the *magnitude* of the contrast is not established at the 95% level by two sites alone. The robust observation is more basic — the two boreholes require *different deep temperature gradients*. The published global Hayne et al. (2017) value, applied unchanged, runs warm at both sites but by very different amounts (+1.0 K mean at A15, +2.9 K at A17; Table 2); the value that zeroes the A17 residual (11.23) over-corrects A15 to −0.50 K, and the value that zeroes A15 (4.86) still leaves A17 warm by +1.63 K.

The deep gradient is set by the ratio $Q_b/K(T,z)$, however, so a steeper gradient at A17 can be produced by a higher K_d *or* by a higher basal heat flux Q_b. The basal fluxes are themselves uncertain — the Saito et al. (2007); Nagihara et al. (2018) reanalyses revise the original Langseth et al. (1976) A15 value downward by ∼30% — so "different K_d" and "different Q_b" are competing explanations that a single-borehole gradient cannot separate. The next subsection tests that competition directly; it is the central interpretive question of this study.

4.2 A uniform K_d with site-specific Q_b is disfavoured but not excluded

Whether the HFE record *requires* an inter-site conductivity contrast, or merely tolerates one, is decided by comparing three models fit to the same pooled deep-sensor set (Table 4): M1, per-site K_d with Q_b fixed at the published values (the retrieval of §3.5); M2, a single *shared* K_d with the per-site Q_b free within the published Langseth et al. (1976); Saito et al. (2007); Nagihara et al. (2018) envelopes; and M3, the strict null of a shared K_d with Q_b also fixed. Free-parameter counts are 2, 3 and 1 respectively, and the small-sample Akaike criterion penalises them accordingly.

The variable-conductivity model M1 is preferred, but not decisively: it improves on the uniform-K_d alternatives by $\Delta\mathrm{AICc} \approx 5$ — enough to disfavour them ($\Delta\mathrm{AICc} > 4$ is conventionally "considerably less support"), but short of the $\Delta\mathrm{AICc} \gtrsim 10$ that would constitute decisive rejection. Two further points temper the uniform-K_d model M2. First, it reaches its best fit only by driving the two basal fluxes to *opposite extremes* of their published ranges ($Q_{b,\mathrm{A15}}$ to the top, 25, and $Q_{b,\mathrm{A17}}$ to the bottom, $10\,\mathrm{mW/m^2}$) — the least likely configuration within the envelope, since no independent evidence favours such an anti-correlated pair. Second, even with that freedom its pooled RMSE (0.59 K) is worse than M1's (0.56 K). We therefore conclude that the HFE record *favours* an inter-site difference in deep conductivity, but does not establish it beyond an alternative in which the conductivity is uniform and the two basal heat fluxes differ. This is the honest limit of what two boreholes can deliver, and we carry it forward as the central caveat on the result.

Table 4. Pooled deep-sensor model comparison ($N = 23$: 7 at A15, 16 at A17). M1 retrieves a per-site K_d; M2 shares one K_d and frees the per-site Q_b within the published envelope; M3 fixes both. k is the free-parameter count; ΔAICc is relative to the best model. K_d in $\mathrm{mW\,m^{-1}\,K^{-1}}$, Q_b in $\mathrm{mW\,m^{-2}}$.

Model	K_d (A15 / A17)	Q_b (A15 / A17)	RMSE (K)	ΔAICc
M1 variable K_d, Q_b fixed	5.0 / 11.0	21 / 15 (fixed)	**0.56**	**0.0**
M2 uniform K_d, Q_b free	6.5 (shared)	25 / 10 (fitted)	0.59	+5.1
M3 uniform K_d, Q_b fixed	13.0 (shared)	21 / 15 (fixed)	0.67	+5.7

4.3 The retrieved absolute K_d values are degeneracy-limited

The Q_b/K_d degeneracy that underlies the model comparison above also bounds how precisely the *absolute* per-site conductivities can be quoted. At steady state the deep gradient $|\partial_z T| = Q_b/K_d$ is invariant under any rescaling $(K_d, Q_b) \to (\alpha K_d, \alpha Q_b)$: a single borehole cannot tell those two products apart. The degeneracy is visible in the joint posteriors (Fig. 8a,b) and mapped in Fig. 6a. With the canonical $Q_b^{\mathrm{A17}} = 15\,\mathrm{mW/m^2}$ (Langseth et al., 1976; Nagihara et al., 2018) we recover $K_d = 11.23$;

with the lower $\sim 10\,\mathrm{mW/m^2}$ Saito et al. (2007); Nagihara et al. (2018) value it rescales *downward* to ~ 7.7, and with the upper end ~ 18 it rises to $\sim 13.8\,\mathrm{mW/(m\,K)}$. The absolute A17 number therefore carries a $\sim\pm 40\%$ envelope set entirely by the basal-flux debate; the absolute A15 value carries a comparable envelope. These per-site numbers should accordingly be read as *conditional* on the adopted Q_b, not as free-standing measurements. What the data constrain tightly at each borehole is the product Q_b/K_d — equivalently the deep gradient — and it is on that better-constrained quantity, not on K_d in isolation, that the inter-site comparison of §4.2 rests.

4.4 If the contrast is a conductivity effect, its size is physically plausible

If the preferred interpretation holds and the inter-site difference is genuinely in K_d, a ratio $K_d^{\mathrm{A17}}/K_d^{\mathrm{A15}} \approx 2$ is large but not implausible for two compositionally distinct sites — though we stress this is a consistency argument, not evidence for the conductivity interpretation over the basal-flux one. Two known mechanisms each act in the right direction. Apollo cores span bulk porosities $\phi \sim 0.30$–0.45 (Mitchell et al., 1973; Carrier et al., 1991); an effective-medium scaling $K_{\mathrm{eff}} \propto (1-\phi)^2$ (Wood, 2020) gives a $\sim 1.5\times$ change for a porosity difference at the upper end of that range. Solid lunar basalt is also $\sim 30\%$ more conductive than solid highland anorthosite (Horai and Susaki, 1972): Apollo 15 samples a mixed mare/highland regolith at the Imbrium–Apennine boundary (Korotev, 2005; Wieczorek et al., 2006), whereas Apollo 17 sits in a basaltic embayment within the Taurus–Littrow highlands (Crawford, 2015). A factor ~ 2 is thus within reach of ordinary regolith variability, but the present two-borehole data neither isolate which mechanism would dominate nor, as §4.2 showed, exclude the competing basal-flux explanation.

4.5 Comparison with prior estimates

The Martínez and Siegler (2021) model is calibrated to orbital Diviner brightness temperatures rather than to the in-situ HFE record, so it provides a genuinely independent comparison; evaluated at the two sites its effective deep conductivity is ~ 7–$8\,\mathrm{mW/(m\,K)}$ (temperature-dependent), intermediate between our A15 and A17 retrievals. It is correspondingly too conductive for A15 and not conductive enough for A17, leaving the deep record warm at both sites (§4.1); a single globally-calibrated relation cannot match the two sites at once. Direct laboratory measurements on Apollo soils (Cremers and Birkebak, 1971; Horai, 1981; Hemingway et al., 1973) give ~ 0.7–$2\,\mathrm{mW/(m\,K)}$, factors of ~ 3 (A15) and ~ 7 (A17) below our in-situ values — as expected, since lab measurements lose the m-scale grain-contact networks and radiative pathways that dominate the in-situ response. Our values bracket the orbital Vasavada et al. (2012) estimate of $\sim 7\,\mathrm{mW/(m\,K)}$, with A15 just below and A17 above.

4.6 Why this matters for polar volatiles and instrument design

Polar volatile-stability calculations and buried-instrument thermal budgets are built on a globally-uniform deep thermal state — a single K_d *and* a single Q_b. The two HFE boreholes are inconsistent with that assumption: whichever of the two variables carries the difference, the deep temperature gradient is not the same at the two sites, and downstream analyses inherit a comparable uncertainty. To illustrate the magnitude, integrating the Hayne $K(T,z)$ profile under polar conditions (Fig. 9) gives a cold-trap ice-stability depth that grows from $\sim 9\,\mathrm{m}$ for the gradient implied by the Hayne et al. (2017) global value to $\sim 29\,\mathrm{m}$ for the steeper gradient retrieved at A17 — a $3\times$ shift in the depth of recoverable polar water ice. The point is not the specific number, which is conditional on the conductivity interpretation, but that a factor-~ 2 inter-site spread in the deep thermal state maps into a several-fold spread in a quantity of direct exploration interest. Future mapping efforts should therefore treat the deep regolith thermal state as spatially variable rather than globally uniform.

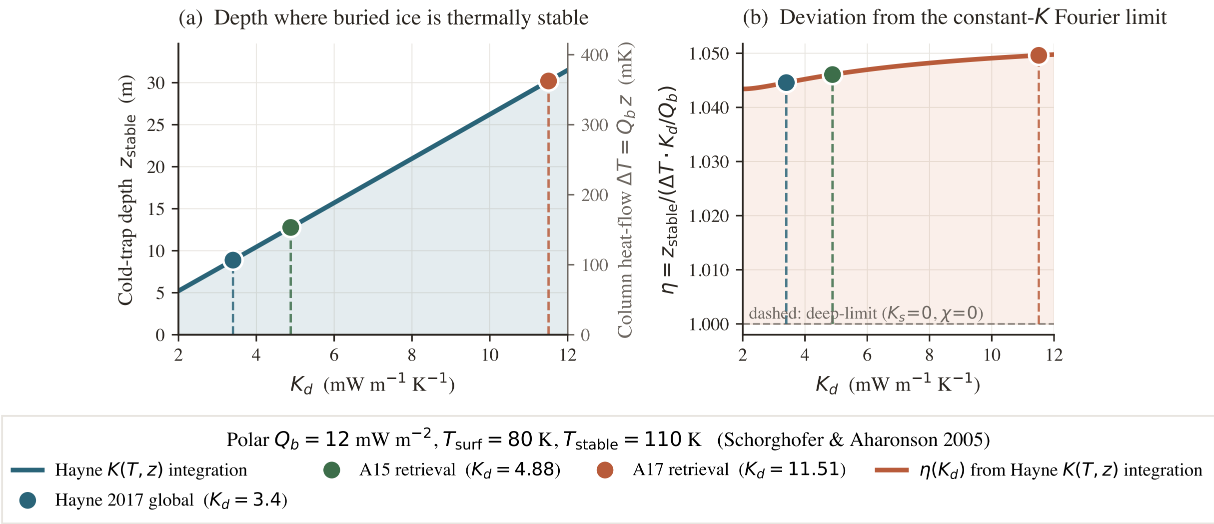

Figure 9. Polar cold-trap depth vs. K_d (Schorghofer-2005-style estimate at $T_{\text{surf}} = 80\,\text{K}$, $Q_b = 12\,\text{mW/m}^2$). (a) z_{stable} from RK4 forward-integration of $dT/dz = Q_b/K(T, z)$ through the Hayne $K(T, z)$ profile until $T = T_{\text{stable}} = 110\,\text{K}$; the right axis recasts the same depth as the column heat-flow drop $\Delta T = Q_b\, z$. The curve is close to linear because the Q_b/K_d term dominates. (b) Ratio $\eta = z_{\text{stable}}/[(\Delta T)\, K_d/Q_b]$ relative to the constant-K Fourier limit; $\eta \approx 1.04$–1.05 resolves the mild ($\sim 5\%$) departure from linearity from the shallow-K_s layer and the radiative term.

4.7 Existing globally-calibrated conductivity models miss A17

With the per-site retrieved K_d^*, the Hayne et al. (2017) form reproduces the deep gradient at both A15 and A17 (Fig. 10), and the this-work 3-layer model does likewise — the retrieval is a property of the data, not of a particular $K(z)$ shape. The two *parameter-free* published predictions, by contrast, both miss. The Hayne et al. (2017) global value runs warm at both sites (Table 2). The genuine Martínez and Siegler (2021) temperature–density model, evaluated with no free parameter, is also warm at both sites but by unequal amounts — a mean deep-sensor residual of $+0.6\,\text{K}$ at A15 and $+2.3\,\text{K}$ at A17 (deep RMSE 1.1 and 2.4 K; Fig. 10c,d) — fitting A15 tolerably while missing A17 by more than 2 K. Both of these models carry a globally-uniform conductivity *and* an implicit globally representative thermal structure; under that joint assumption neither reproduces the A17 deep record. This is consistent with §4.2: the A17 borehole requires either a higher deep conductivity or a higher basal flux than a single globally calibrated thermal model provides. We emphasise that this does not by itself isolate the conductivity as the variable — a globally calibrated model could equally be failing at A17 because the basal flux there departs from the global mean.

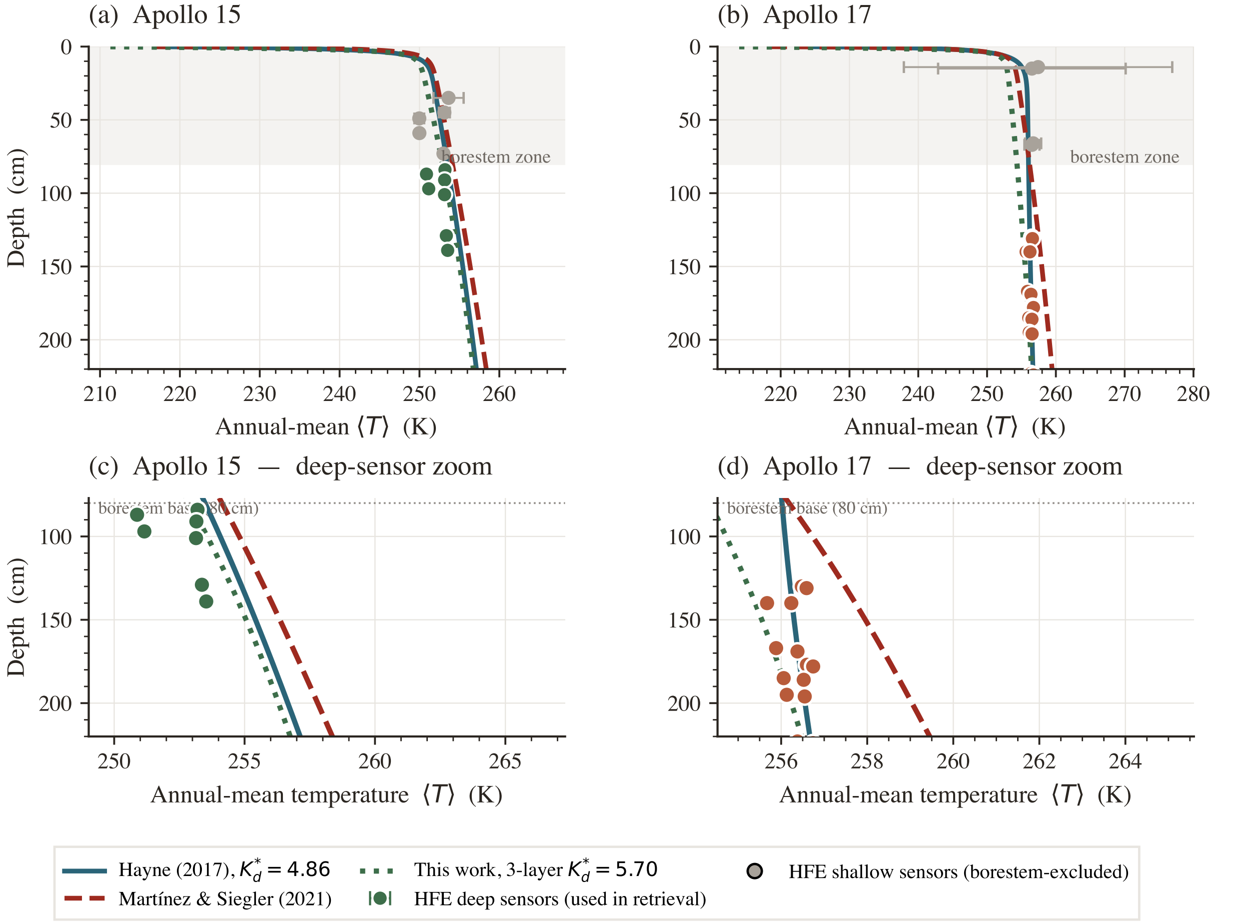

Figure 10. Annual-mean $T(z)$: three conductivity models against the Apollo HFE deep sensors. Solid teal: Hayne et al. (2017) at the per-site retrieved K_d^*; dashed red: parameter-free Martínez and Siegler (2021); dotted green: this-work 3-layer at its per-site retrieved K_d^*. Top panels: full profile; bottom panels: deep-sensor zoom. Coloured markers: HFE deep sensors (used in retrieval); grey markers: borestem-affected shallow sensors (excluded).

4.8 Limitations and caveats

Several caveats should accompany any use of the result. *(i)* With only two in-situ sites, the inter-site difference is not sufficient to establish *regional* heterogeneity across the lunar surface; distinguishing genuine regional variability from local-scale heterogeneity between two specific borehole locations requires additional landed thermometry. *(ii)* The deep gradient constrains the product Q_b/K_d, not K_d alone (§4.2); the data favour, but do not establish, a conductivity contrast over a uniform conductivity with differing basal flux. *(iii)* The contrast *magnitude* is not significant at the 95% level ($\Delta K_d^* = 6.6$, bootstrap CI $[-3.2, 16.6]$); only the requirement of different deep gradients, and the model-comparison preference, are robust. *(iv)* The retrieval assumes the Hayne et al. (2017) functional form for $K(T, z)$; we test sensitivity to that shape with a discrete 3-layer profile (§3.5), but other piecewise parameterisations are not exhausted. *(v)* The diurnal-amplitude depth profile (Fig. 3) is used only as a borestem diagnostic and not as a frequency-domain validation, because below 80 cm both models predict amplitudes well below the 1971–1977 digitisation noise floor.

5 Conclusions

We set out to test how faithfully a globally-averaged deep regolith conductivity represents an individual location, using the only data that can answer the question at depth: the restored Apollo 15 and 17 Heat-Flow Experiment record. A single-parameter retrieval, performed within the Hayne et al. (2017) functional form and propagated through a non-parametric bootstrap, gives $K^*_{d,\mathrm{A15}} = 4.86^{+1.12}_{-0.54}$ and $K^*_{d,\mathrm{A17}} = 11.23^{+8.12}_{-0.77}\,\mathrm{mW/(m\,K)}$ at the published basal heat flux. The two values bracket the global figure rather than reproduce it, and an independent three-layer profile built from the Apollo core measurements returns the same ordering — so the per-site departure is a property of the data, not of the assumed functional form.

These absolute values are conditional on the adopted Q_b: the deep gradient constrains the ratio Q_b/K_d, not K_d alone. We addressed the resulting ambiguity with a pooled three-model comparison. A variable-K_d model is preferred over a uniform-K_d model with site-specific Q_b, though by a modest ΔAICc ≈ 5, and the uniform-K_d fit succeeds only when the two basal fluxes are driven to opposite extremes of their published ranges. The record therefore favours a genuine inter-site conductivity contrast of roughly a factor of two, without excluding a heat-flux contrast of comparable effect; the underlying observation — that the two boreholes demand distinctly different deep gradients — is itself robust to the borestem exclusion depth, the three-layer profile, a joint K_d–H fit, and independent Diviner surface-temperature closure.

The practical outcome is the error budget this study was designed to deliver. A deep conductivity taken from a Moon-wide average and applied at a specific site should be expected to differ from the local value by a factor of order two, with the residual ambiguity between conductivity and basal flux folded into that figure. This is a quantity to be carried explicitly — in polar volatile-stability calculations, in buried-instrument heat budgets, and in any site-resolved thermal model — rather than left as a silent assumption of uniformity. Narrowing it further is no longer a modelling problem but a measurement one: it requires deep temperatures from more than two locations, for which forthcoming landed heat-flow instruments (e.g. Chang'E and ChaSTE-class experiments) are the natural next data.

Acknowledgments

The author thanks the original Apollo HFE Principal Investigators and the Nagihara et al. (2018) team for restoring the 1975–1977 portion of the 1971–1977 record from the original tapes (the 1971–1975 portion was retained from the original PI processing), without which this work would not have been possible. The author declares no funding sources for this work.

Open Research

Availability Statement. The restored Apollo 15 and 17 Heat-Flow Experiment temperature record (1971–1977) analysed here is openly available as the supporting information of Nagihara et al. (2018) (`https://doi.org/10.1029/2018JE005579`). The Diviner Lunar Radiometer Global Cumulative Product used for the out-of-sample surface-temperature closure (Williams et al., 2017) is available from the NASA Planetary Data System Geosciences Node (`https://pds-geosciences.wustl.edu`; LRO-L-DLRE-5-GCP-V1.0, `https://doi.org/10.17189/1520650`). Surface and subsurface geometry were computed with the NASA SPICE Toolkit (Acton, 1996; Acton et al., 2018) and the JPL DE440 ephemeris. The 1-D Crank–Nicolson heat solver, the ephemeris-driven solar-forcing routines, the K_d retrieval and bootstrap code, the validation diagnostics, and the data and scripts needed to reproduce every figure and table in this manuscript and its supporting information are archived at `https://github.com/rp3gregorio/Lunar-V2`; a Zenodo-archived release with a citable DOI will be issued at the time of acceptance and added to the reference list.

Conflict of Interest Disclosure

The author declares there are no conflicts of interest for this manuscript.

Nomenclature

Symbols.

Symbol	Definition	Value / units	Source
K_d	deep regolith thermal conductivity (retrieved)	4.86, 11.23 $\mathrm{mW\,m^{-1}\,K^{-1}}$	this work
K_s	surface regolith thermal conductivity	7.4×10^{-4} $\mathrm{W\,m^{-1}\,K^{-1}}$	Hayne et al. (2017)
H	e-folding depth of $K(z)$ transition	0.06 m	Hayne et al. (2017)
χ	radiative conductivity multiplier coefficient	2.7	Hayne et al. (2017)
T_{ref}	normalisation temperature for radiative term	350 K	Hayne et al. (2017)
ρ_s, ρ_d	surface / deep regolith bulk density	1100, 1800 $\mathrm{kg\,m^{-3}}$	Hayne et al. (2017)
$c_p(T)$	specific heat (4th-order polynomial in T)	$\mathrm{J\,kg^{-1}\,K^{-1}}$	Hemingway et al. (1973); Hayne et al. (2017)
Q_b	basal heat flux	21 (A15), 15 (A17) $\mathrm{mW\,m^{-2}}$	Langseth et al. (1976); Nagihara et al. (2018)
S_0	solar constant (irradiance at 1 AU)	1361 $\mathrm{W\,m^{-2}}$	Kopp and Lean (2011)
A, ε	broadband Bond albedo, emissivity	site-specific; 0.95	Vasavada et al. (2012); Bandfield et al. (2015)
σ	Stefan–Boltzmann constant	5.6704×10^{-8} $\mathrm{W\,m^{-2}\,K^{-4}}$	CODATA 2018
$\theta_\odot(t)$	solar zenith angle at site	rad	SPICE/DE440
$F_\odot(t)$	incident solar irradiance at surface	$\mathrm{W\,m^{-2}}$	this work, eq. 1
$T(z,t)$, T_s	subsurface and skin ($z=0$) temperature	K	this work
z	depth coordinate (positive downward)	m	—
80 cm	borestem-exclusion depth (*a priori* cut)	80 cm	this work, §3.3
δ	diurnal thermal skin depth	∼3–5 cm	Hayne et al. (2017)
z_{stable}	cold-trap ice-stability depth	m	Schorghofer and Aharonson (2005)
ΔK_d	inter-site contrast $K_d^{\mathrm{A17}} - K_d^{\mathrm{A15}}$	6.63 $\mathrm{mW\,m^{-1}\,K^{-1}}$	this work
RMSE	deep-sensor root-mean-square residual	K	this work
N	number of deep sensors	7 (A15), 16 (A17)	Nagihara et al. (2018)
N_{boot}	non-parametric bootstrap resamples	1500	Efron and Tibshirani (1993)
p	bootstrap proportion below null contrast	0.05	this work

Subscripts.

s	surface (regolith, skin)
d	deep regolith
b	basal (lower BC)
ref	reference (normalisation)
$\odot$	solar
eq	stability-window equilibrium
boot	bootstrap quantity

Superscripts.

$*$	best-fit / retrieved (e.g. K_d^*)
A15,A17	per-site qualifier
ref	published reference value
HFE	Apollo HFE-measured
$\langle\cdot\rangle$	time-average over stability window

Acronyms.

HFE	Heat-Flow Experiment (Apollo 15 and 17 buried thermometers)
TG / TR	gradient-bridge / ring-bridge sensor (HFE sensor types)
LST	local solar time
Diviner GCP	Diviner Lunar Radiometer Global Cumulative Product
MCMC	Markov-chain Monte Carlo
BCa	bias-corrected accelerated bootstrap
SPICE	NASA ancillary information system for planetary geometry
DE440	JPL planetary and lunar ephemeris release 440
SI	Supporting Information

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